DSAI4901 · CAPSTONE PROJECT PROPOSAL

Paper-Distillation PPT Agent

From generating to verifying

A slide generation system built around a measurable definition of deck usability

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October 2026

38
papers reviewed

8
functional gaps

1
research direction

Literature review: from 367 papers to 38

367 arXiv candidates

69 after relevance screening

38 read in depth (33 agents + 5 education)

18 with author-released code

18/38 release author code (each link verified by HTTP); the rest are paper-only or anonymous submissions

WHAT I ACTUALLY DID

- 12 arXiv queries across slide, PPT, poster and deck
- Relevance screening to remove unrelated work
- Full-text keyword scan of all 38 papers
- GitHub links extracted from each PDF, then verified by HTTP

Four things they already do well

First establish what is solved — this decides what not to rebuild

01

Generation & layout

Decks already look convincing: stable layout, consistent styling.

Zheng et al. (2025); Vats et al. (2026)

02

Native editable output

Work has moved into the .pptx object layer, preserving editability.

Jung et al. (2025); Zheng et al. (2026b)

03

Evaluation methods

From holistic scores to task-based tests and rubric gates.

Zhang et al. (2025); Jang et al. (2026)

04

Multimodal & narrative

Figures and tables enter the layout; story structure is modelled.

Liang et al. (2025); Ozden et al. (2026a)

Conclusion: generation is commoditised. One more generator is not a contribution.

Decomposing a PPT agent into 8 functions

A precise gap needs a function map — the next four slides walk through it

① **Parsing** → ② **Content** → ③ **Narrative** → ④ **Layout** → ⑤ **Output** → ⑥ **Inspection**

⑦ **Speaker notes** — Cross-stage: turn content into speakable pacing

⑧ **Provenance** — Cross-stage: every sentence needs a source

Next four slides: two gaps per slide — symptom · evidence · consequence

Gap 1 | Parsing and content selection

① PARSING

Figures and captions split apart

SYMPTOMS

- figure and caption are detached, so image and text no longer correspond
- two-column papers are read in the wrong order
- tables are flattened into a wall of text

EVIDENCE · APA

Parsing is treated as preprocessing and figure–text alignment is not enforced (Pang et al., 2025; Fu et al., 2021).

CONSEQUENCE

A corrupted input is inherited by every later step, and the error is invisible in the final deck.

② CONTENT SELECTION

Even allocation, no emphasis

SYMPTOMS

- one slide per section, regardless of how much it matters
- the key result is dropped while background survives
- method details are pushed into the introduction slide

EVIDENCE · APA

Most methods concentrate on text selection and ignore the link between space and importance (Liang et al., 2025; Chen et al., 2026).

CONSEQUENCE

The deck is complete but has no emphasis; the audience keeps only the title.

Gap 2 | Narrative and layout

③ NARRATIVE

A table of contents, not a talk

SYMPTOMS

- section-by-section summarising reproduces the paper's order
- ten slides of background, two of results
- no motivation → problem → insight → evidence arc

EVIDENCE · APA

Source text is summarised straight into slides (Ozden et al., 2026a); explicit narrative planning remains rare (Vats et al., 2026).

CONSEQUENCE

The speaker reads a table of contents aloud and the audience cannot build causal understanding.

④ LAYOUT

Overflow, overlap, clipping

SYMPTOMS

- text runs past the slide edge; elements overlap
- fonts too small and contrast too low once projected
- figures cropped, every slide a wall of bullets

EVIDENCE · APA

Local failures include overflow, overlap, clipping and off-canvas placement (Tian et al., 2026); models understand content yet fail at layout (Huang et al., 2025).

CONSEQUENCE

The most visible failure: text clipped and figures covered during the talk, so the audience loses trust.

Gap 3 | Output and inspection

⑤ OUTPUT

Looks usable, cannot be edited

SYMPTOMS

- output is an image or HTML, not editable in PowerPoint
- even .pptx output is all text boxes — one edit breaks the layout
- charts are rasterised and the data is gone

EVIDENCE · APA

Agents frequently rasterise editable content or break the layout (Zheng et al., 2026b); only an object-model route preserves styling (Jung et al., 2025).

CONSEQUENCE

Users get an artefact they cannot modify, so it is unusable and the tool is abandoned.

⑥ INSPECTION

One-shot feedback, no attribution

SYMPTOMS

- the whole deck is rewritten and reviewed once
- errors cannot be localised to an element
- repair can make things worse, with no convergence guarantee

EVIDENCE · APA

A monolithic one-version-one-feedback loop makes local errors hard to attribute (Tian et al., 2026); reflection must be grounded in rendered state (Zheng et al., 2026a).

CONSEQUENCE

Iteration becomes a gamble: each pass may degrade the deck and nobody can say why.

Gap 4 | Speaker notes and provenance

⑦ SPEAKER NOTES

Notes are the bullets again

SYMPTOMS

- speaker notes duplicate the text on the slide
- no time budget — no idea how many slides fit three minutes
- no transitions, so slides do not connect

EVIDENCE · APA

Delivery is under-optimised (Yang et al., 2026); only 5 of 38 papers mention speaker notes, none as a research focus.

CONSEQUENCE

The speaker reads the slide while the audience reads and listens at once.

⑧ PROVENANCE

Writes the words, cannot cite them

SYMPTOMS

- bullets carry no source, so there is no path back to a passage
- related work is presented as 'we propose'
- figures and tables misnumbered; numbers attached to the wrong claim

EVIDENCE · APA

Retrieval improves reliability but per-sentence provenance is not exposed (Xie et al., 2025); global intent is kept while sentence-level anchors are missing (Ozden et al., 2026a).

CONSEQUENCE

A wrong number goes unnoticed; in academic settings one misattributed claim destroys credibility.

A single failure is amplified downstream



Every upstream error is amplified downstream — and cannot be traced back

Audience

Cannot follow or remember; sees clipped text live

→ Attention lost

Speaker

Will not trust it; cannot answer 'where does this number come from?'

→ Trust collapses

Field

Cannot compare systems, because there is no shared measure

→ Progress stalls

Measurement now exists (SlideAudit, DECKBench). What is missing is using it as a control signal — and measuring what it buys.

Why this direction and not the others

Four alternatives were considered seriously; each fails on a specific constraint

Direction considered	Why it is attractive	Why we are not doing it
Reducing hallucination	Compelling story; high novelty	Hallucination is a property of the base model. A student can only wrap RAG around it, which fixes fabrication but not over-claiming or misattribution. Ground truth needs sentence-level human checking: 300 bullets × 2 min.
A finer taxonomy	Clear definition; easy to write	SlideAudit already published 27 categories over 2,400 slides, built with design experts and trained annotators. A solo project cannot match that.
A multi-agent architecture	Sounds advanced	Per-module ablation becomes near-impossible, and the first defence question is which agent actually contributes.
Cost optimisation	Easy to quantify	Cost is the denominator, not the goal. Pure engineering tuning is not a research contribution.

The intersection we chose: mechanically decidable × per-class attributable × feasible for one person.

The eight works this direction builds on

Each one plays a specific role — none is simply 'related work'

SlideAudit
Label space
six overlapping categories, re-used verbatim

DECKBench
Contrast
reference-aware scoring; needs a target deck

PPTArena / PPTPilot
Baseline ③
the loop we run as a controlled ablation

ReDeck
Loop design
one edit, one observation

PPTAgent
Baseline ②
strongest reproducible generator

Paper2Poster
Evaluation idea
answer questions using only the artefact

PosterMELD
Concept source
print-ready rate → slide-ready rate

SLIDEFORGE
Representation
pptx object layer instead of pixels

My direction: paper distillation, not lecture notes

Turn computable layout violations into an in-loop control signal, and measure what it buys

IN SCOPE

- single-column English papers, 8–10 in total
- 10–12 slides per deck, editable .pptx output
- six violation classes, per-class reporting
- speaker notes: 2–3 lines per slide plus a time budget
- 20-participant pairwise human study

OUT OF SCOPE

- lecture notes and teaching material
- multi-column or scanned papers
- Chinese / bilingual output
- hallucination measurement
- multi-agent architecture as a selling point

RQ1	Can layout usability be measured without a reference deck?
RQ2	Does a rule-based verifier beat a VLM judge in the same loop?
RQ3	Which violation classes improve, and which stay flat?
RQ4	What does each improvement cost?
RQ5	Can every bullet carry a resolvable source anchor?

After the main line — provenance module (C5): per-bullet source anchor, 100 bullets, about four hours of annotation.

Five contributions, in order of weight

C1	**A reference-free, object-level measure of usability**	SRR, deck-ready and per-gate pass rates, computed on the native .pptx — no ground-truth deck required.
C2	**Measurement turned into a control signal**	detect → edit → re-render → re-measure; the verifier is isolated as the only variable.
C3	**Per-class attribution**	which classes repair well, which do not, and why. This is the paper.
C4	**A rerunnable research artefact**	150 annotated slides (40 calibration + 110 test), rulebook, detectors, scripts.
C5	**Provenance anchors (secondary)**	every bullet resolves to a passage; reported as coverage and accuracy.

C1 is the ruler, C2 is the hand, C3 is the finding. C4 makes it reproducible; C5 is cut first if time runs out.

Operational definition: slide-ready rate

SRR = slides passing every gate / slides; deck-ready is a separate binary. Both are always reported.

Gate	Rule	Threshold	Anchor	Swept over
G1 Containment	Every shape box inside the slide; no image cropped	0 pt	binary rule	—
G2 Overlap	Text–text intersection ÷ area of smaller box	> 5%	empirical	2 / 5 / 10%
G3 Font size	Body and titles, read from run properties	18 / 28 pt	design guidance	16 / 18 / 20
G4 Contrast	WCAG 2.2 AA ratio, fill vs background	4.5:1 (3:1 large)	published standard	—
G5 Text fit	Rendered text height ÷ box height	≤ 0.95	renderer tolerance	0.90 / 1.00
G6 Density	Characters per slide; bullets per block	≤ 140; ≤ 6	empirical	120 / 160

Calibration: sweep each threshold, keep the value that maximises agreement with human labels.

Aggregation: SRR is slide-level and continuous; deck-ready = every slide passes (9 of 10 is NOT deck-ready).

Positioning: SlideAudit supplies the label space; DECKBench scores against references; SRR is reference-free — a control signal, not a score.

Evidence plan: agreement, power, fallbacks

The grade depends on whether a sceptic can rerun these three numbers

DETECTOR VALIDATION

- 150 slides = 40 calibration + 110 held-out, stratified across all three systems
- 2 independent annotators; rulebook training plus 10 calibration slides
- Cohen's kappa >= 0.70 per class, adjudicated by a third pass
- precision / recall / F1 per class, never an aggregate only
- if per-class F1 < 0.70 on calibration, revise the rule and re-freeze

HUMAN STUDY

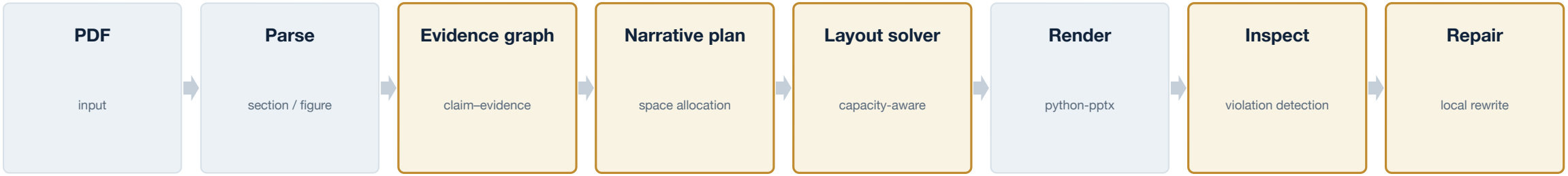
- within-subject: 20 raters x 4 papers, all three systems on the same paper
- one primary contrast pre-registered: ours vs the VLM-judge loop
- Holm–Bonferroni across the remaining two comparisons
- n = 20 detects d about 0.65–0.70 at 80% power
- raters blind to condition order; papers rotated

IF RECRUITMENT FALLS SHORT OF 20

3 expert raters with ICC(2,k) · expand automated QA from 10 to 25 items · report posterior probability of superiority · label the result exploratory

System architecture

Gold = new contributions; the rest reuses validated practice



⑨ Speaker notes — 2–3 lines per slide plus a time budget, written into .pptx notes; generated once, outside the control loop.

Restrict the output space

the model writes structured content only; geometry belongs to the solver

One element at a time

local rewrites with immediate verification make errors attributable

Every judgement has a number

no 'looks fine' — only measured violations and ratios

The output is not an image but an editable .pptx: layout comes from constraint solving, content from the evidence graph.

What the repair actually does

One decision procedure per violation class: cheapest action first, with a guardrail that forbids adding content

Violation	What we do, in order	Guardrail
Text out of bounds / clipped	shrink font one step → reduce box margin → move the box inward → split the slide	never below 14 pt
Overlapping text	move the lower-priority element → resize the box → reflow into two columns	keep image z-order
Text over a busy background	add a scrim behind the text → move the text → change the text colour	contrast must reach 4.5:1
Text too long	delete words (never add) → split across two slides	no new claims, no number changes
Too dense	split one slide into two	preserve claim order

Order matters: the cheapest reversible action is tried first, so the cost of a repair is predictable.

Three ways to build the loop — and the order we take

All three are feasible; they differ in cost, risk and how early they produce a number

Route	What it does	Cost / risk	When
A Deterministic repair	shrink font → move → resize → split slide, no LLM call	cheapest, fully reproducible; limited reach	W5–7
B Detector-conditioned rewrite	violations become constraints; the LLM rewrites one slide's structured content	1 call per slide per pass; content drift	W8–11
C Hybrid two-loop	inner loop A + a VLM pass every third iteration	highest complexity; hardest to debug	W12–14

A alone already yields a reportable result. B is added only if A converges; C exists only as the comparison arm.

How the claims are tested

Three tracks in parallel, avoiding the circularity of an LLM judging an LLM

Track	Metric	Sample	Why it is needed
Automatic (geometry)	Violation count, SRR, deck-ready	150 slides (40 + 110)	deterministic and rerunnable
Task-based (QA)	answer accuracy from the deck	10 questions x 8 papers	can a reader extract correct facts?
Coverage	key claims that survive into the deck	5-8 claims per paper	required by the brief
Human	pairwise preference, speakability	20 participants	design has no ground truth

BASELINES — EACH ANSWERS A DIFFERENT QUESTION

1 **Naive single-pass** — does inspection add anything at all?

2 **PPTAgent** — versus the strongest reproducible generator

3 **VLM-judge repair loop** — computable rules versus subjective judgement

3 is the primary claim: same loop, same budget, only the verifier changes.

ABLATION

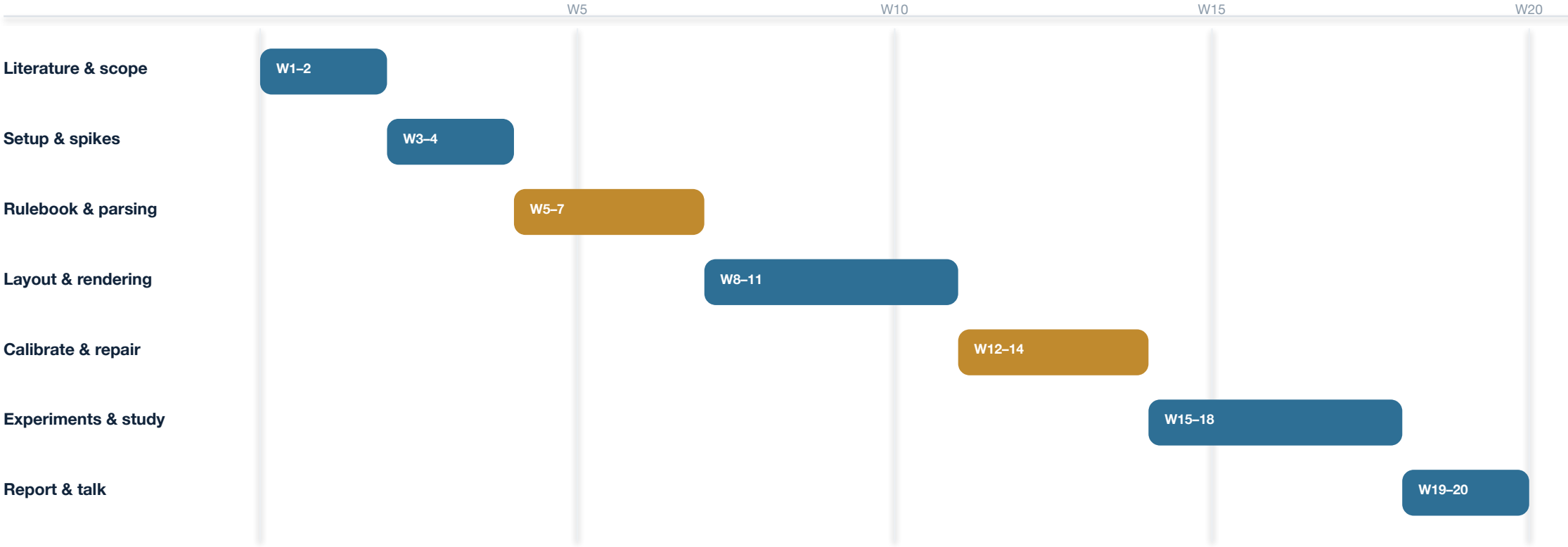
Remove anchors
Remove inspection
Remove repair
→ per-module attribution

Report per class,
including where repair fails.

Inspection starts in W5, ahead of the brief's stage 5 - the metric must be validated before the loop that depends on it.

20-week timeline

Four parallel workstreams; if recruitment is short, fall back to expert rating plus expanded QA



2026-11-05

Proposal submission

2027-01-07

Interim Report

2027-04-08

Final Report + Turnitin + AI Report

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